

Kirchhoff	Robin		BDF	3
57.4741 h		4	51.0922 h	

关键字：

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一、 问题重述

二、 问题分析

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“

”

$$1/R^2$$

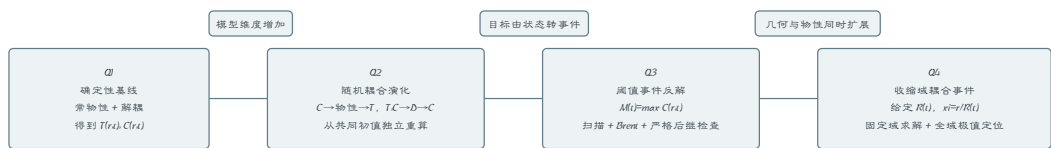
$$1/R$$

Kirchhoff

Robin

BDF

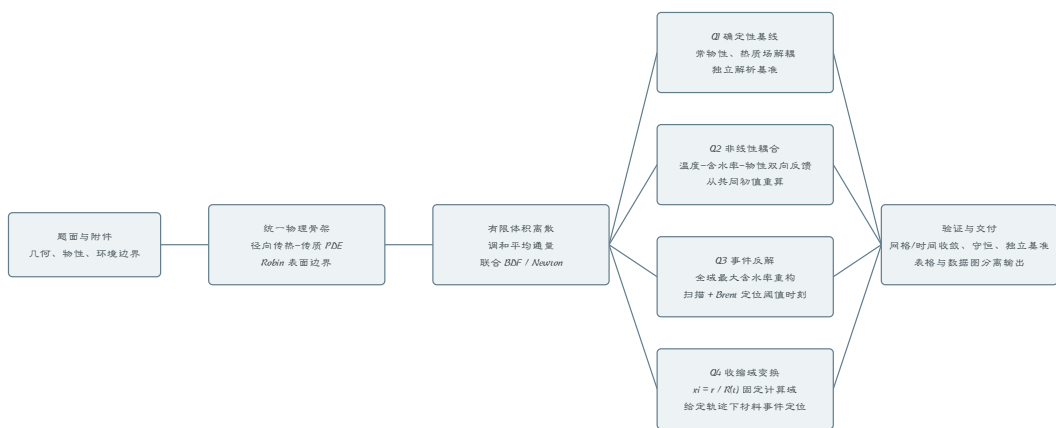
[[1, 2]]



1:

1

2



2:

三、 模型假设

1.

2.

Robin

3.

4 h

4.

2

3

4

5.

四、符号说明

主要符号说明

t	s
r, ξ	m
$R(t)$	m
T, C	K kg/kg
ρ, c_p, k, D	SI
h, k_m	W/(m ² K) m/s
T_∞, C_e	K kg/kg
N	

五、问题一的模型建立与求解

5.1 控制方程与边界条件

$T(r, t) \quad C(r, t) \quad R_0 = 0.02 \text{ m}$

$$\rho c_p \frac{\partial T}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left(r k \frac{\partial T}{\partial r} \right), \quad \frac{\partial C}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left(r D(C) \frac{\partial C}{\partial r} \right). \tag{1}$$

$T_r(0, t) = C_r(0, t) = 0$

$-kT_r(R_0, t) = h(T_s - T_\infty), \quad -D(C_s)C_r(R_0, t) = k_m(C_s - C_e). \tag{2}$

$T_\infty, C_e \quad 1 \quad 2 \quad D(C) = 7 \times 10^{-9} \exp(-0.89/C)$
[[2]]

5.2 环形有限体积离散

$[0, R_0] \quad N$

Kirchhoff

$$K(C)=\int_0^C\exp(-0.89/s)\,\mathrm{d}s,$$

$$D(C)C_r=7\times10^{-9}K_r\tag{[4]}$$

Robin

Brent

5.3 时间推进与结果

BDF

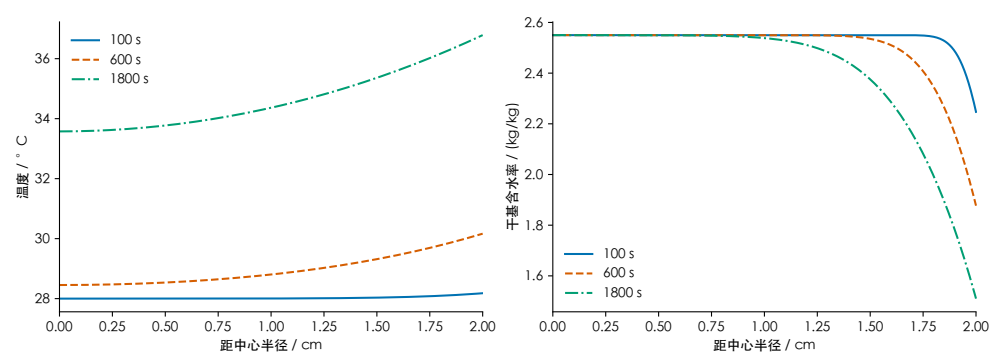
60 s

$N=4096$

10^{-10}

10^{-11}

10 s



3:

		1:		°C		
/s	0 cm	0.5 cm	1 cm	1.5 cm	2 cm	
100	28.0001	28.0003	28.0040	28.0327	28.1801	
300	28.0408	28.0635	28.1514	28.3681	28.8489	
600	28.4533	28.5360	28.8039	29.3159	30.1652	
900	29.3243	29.4583	29.8755	30.6161	31.7304	
1200	30.5427	30.7098	31.2223	32.1126	33.4276	
1500	31.9957	32.1867	32.7660	33.7463	35.1203	
1800	33.5753	33.7720	34.3642	35.3621	36.7856	

		2:		kg/kg		
/s	0 cm	0.5 cm	1 cm	1.5 cm	2 cm	
100	2.5500	2.5500	2.5500	2.5500	2.2470	
300	2.5500	2.5500	2.5500	2.5492	2.0508	
600	2.5500	2.5500	2.5500	2.5353	1.8770	
900	2.5500	2.5500	2.5497	2.5045	1.7547	
1200	2.5500	2.5500	2.5482	2.4646	1.6586	
1500	2.5500	2.5499	2.5445	2.4206	1.5787	
1800	2.5500	2.5497	2.5383	2.3755	1.5102	

1 2 results/q1/table1.csv
 results/q1/table2.csv 3

	2048	4096		1.504×10^{-7} K
2.851×10^{-5} kg/kg				1.537×10^{-7} K
2.950×10^{-9} kg/kg				1.804×10^{-7} K

六、 问题二的模型建立与求解

$$T(r, 0) = 301.15 \text{ K} \quad C(r, 0) = 2.55$$

$$\rho, c_p, k \quad C \quad D(T, C)$$

[[5]]

$$\rho(C)c_p(C)T_t = \frac{1}{r}\partial_r(rk(C)T_r), \quad C_t = \frac{1}{r}\partial_r(rD(T, C)C_r),$$

					Kirchhoff
			Robin		
$N = 1024$		2×10^{-7}		2×10^{-9}	300 s
	3:	3 h		°C	
/h	0 cm	0.5 cm	1 cm	1.5 cm	2 cm
0.5	32.1893	32.3821	32.9660	33.9608	35.4130
1.0	40.3815	40.5535	41.0605	41.8782	42.9976
1.5	45.8465	45.9346	46.1928	46.6049	47.1399
2.0	48.4496	48.4875	48.5979	48.7730	49.0029
2.5	49.4667	49.4789	49.5133	49.5650	49.6607
3.0	49.8494	49.8552	49.8744	49.9100	49.9663

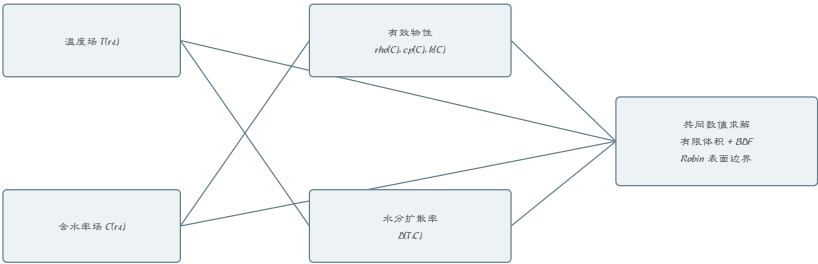
	4:	3 h	kg/kg		
/h	0 cm	0.5 cm	1 cm	1.5 cm	2 cm
0.5	2.5499	2.5489	2.5256	2.3257	1.6486
1.0	2.5257	2.4948	2.3578	2.0230	1.4711
1.5	2.3861	2.3256	2.1344	1.8020	1.3476
2.0	2.1709	2.1084	1.9236	1.6259	1.2311
2.5	1.9567	1.9006	1.7360	1.4720	1.1167
3.0	1.7662	1.7165	1.5702	1.3333	1.0081

3

4

results/q2/table3.csv

results/q2/table4.csv



4:

—

4

3 h

1024

2048

1.163 × 10⁻⁶ K

4.243 × 10⁻⁷ kg/kg

5.912 × 10⁻⁴ K

1.429 × 10⁻⁵ kg/kg

2.665 × 10⁻¹⁵

七、 问题三的模型建立与求解

3

$C_h(r,t)$

$$g(t) = \max_{0 \leq r \leq R_0} C_h(r,t) - 0.15.$$

$g(t)$

600 s

Brent

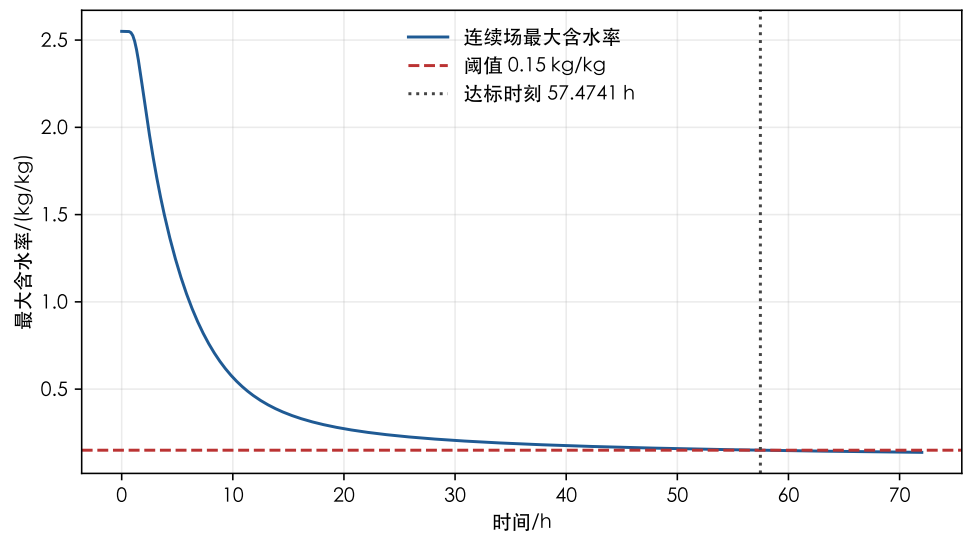
$$t_* = 57.47406473 \text{ h} = 206906.633 \text{ s}.$$

0.1500000000 kg/kg

60 s

57.483333 h

0.1499902825 kg/kg



5:

5:	kg/kg					
/h	0 cm	0.5 cm	1 cm	1.5 cm	2 cm	
6	1.0170	0.9881	0.9015	0.7550	0.5328	
12	0.4562	0.4432	0.4031	0.3297	0.1642	
18	0.2988	0.2916	0.2687	0.2253	0.0845	
24	0.2378	0.2327	0.2167	0.1855	0.0665	
30	0.2058	0.2018	0.1892	0.1641	0.0598	
36	0.1859	0.1825	0.1718	0.1504	0.0566	
42	0.1721	0.1691	0.1597	0.1407	0.0548	
48	0.1618	0.1592	0.1507	0.1334	0.0537	
54	0.1539	0.1514	0.1436	0.1277	0.0530	
57.4741	0.1500	0.1477	0.1402	0.1249	0.0526	

5

results/q3/table5.csv

$N = 256, 512, 1024$

57.47424678 57.47410135 57.47406473 h

-0.198 s

5

八、 问题四的模型建立与求解

2

$R(t)$

$\xi = r/R(t)$

$0 \leq \xi \leq 1$

$v_r = (\dot{R}/R)r$

$U_t = \frac{1}{R(t)^2 \xi} \partial_\xi (\xi D(U, \Theta) U_\xi), \quad \rho(U) c_p(U) \Theta_t = \frac{1}{R(t)^2 \xi} \partial_\xi (\xi k(U) \Theta_\xi).$

$-k \Theta_\xi / R = h(\Theta_s - T_\infty) \quad -DU_\xi / R = k_m(U_s - C_e)$

4

ρ, c_p, k, D

Kirchhoff

Robin

BDF

$M(t) = \max_{0 \leq \xi \leq 1} C_h(\xi, t), \quad t_* = \inf\{t : M(t) \leq 0.15\}.$

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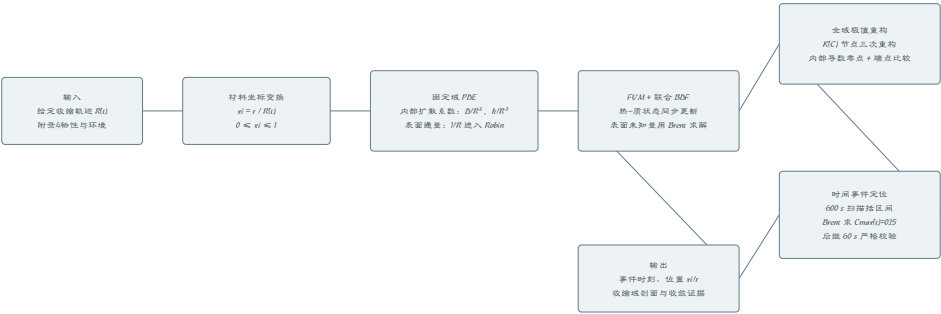
4

$t_* = 51.09216535 \text{ h}$

60 s

51.100000 h

10



7:

/ 3/4

57.47406473	129.84840533	25.24400627	51.09216535 h
-55.493149 h		49.111250 h	-46.526182 h

九、 敏感性分析

				Q3	$N = 512$	$N = 1024$
	0.132 s	Q4	—	4	0.136 s	Q3
Q4	-0.198 s		-0.520 s			
			4 h			Q3
18.28 min	Q4		16.07 min	Q4	PCHIP	
12.75 s		7×10^{-5}				
h	k_m			Sobol		

十、 模型评价与推广

Robin

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附录 A 核心代码

code/q1/-code/q4/

```
for question in Q1, Q2, Q3, Q4:
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```
    BDF      Robin  
    /
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Q3/Q4:      Brent
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